

WANG KANGDI (LERTOON WANG)

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SUMMARY

Audio AI researcher and engineer working at the intersection of **neural audio representation, generative music, and professional music production**. Core contributor and **second author of Qwen-Music**, an industrial-scale music generation system, with hands-on experience in audio quality optimization, music mixing/editing, evaluation, data pipelines, and music agents. **First author of the ϵ ar-VAE series (INTERSPEECH 2026)**, including ϵ ar-VAE2 for high-fidelity stereo music reconstruction with frequency-aware complex-spectral modeling. In parallel, possess **3+ years of stem/live mixing experience**, received professional mixing training at **Modern Sky**, and developed practical DAW/audio tooling. Interested in generative audio, audio representation, DSP, music understanding, and agentic audio systems.

EDUCATION

Heriot-Watt University, Robotics, *B.S.*

2021-09 – 2025-06

Avg score: 83/100, first-class honour.

SKILLS

- **Audio ML**: PyTorch, VAE, GAN, DiT, neural audio representation, music generation, TTS, multimodal audio
- **DSP**: STFT, magnitude/phase modeling, stereo imaging, perceptual audio quality, audio reconstruction
- **Audio Engineering**: Pro Tools, Studio One, Reaper, VST, stem/live mixing
- **Programming**: Python, C++, Lua (ReaScript)

RESEARCH & PROJECTS

ϵ ar-VAE(INTERSPEECH 2026) / ϵ ar-VAE2

2025.07 – Present

- **First author / core author**. Developed neural audio representations for high-fidelity music reconstruction, combining VAE-GAN modeling with perceptually driven supervision and artifact-focused training to recover high-frequency details in complex stereo music.
- ϵ ar-VAE2 further introduces **frequency-aware complex-STFT modeling of magnitude, phase, and stereo structure**, improving continuous audio representation and reconstruction fidelity.
- [Online Demo](#).

Qwen-Music

2026.03 – Present

- **Core contributor / second author** of the Qwen-Music technical report, an industrial-scale unified music generation system. Contributed to decoder-side audio quality, interactive mixing/editing, evaluation, and data-quality pipelines.
- Designed music-specific human evaluation and acceptance criteria, refined annotation/curation workflows, and explored integration of multimodal reasoning, coding/function-calling, and music understanding/editing.
- [Technical Report](#).

CineDub (ACM MM 2026)

2026.02 – 2026.04

- **Co-first author**. Developed a Diffusion Transformer based video dubbing model under weak conditioning, contributing to global temporal conditioning, negative conditioning, training, and latent-decoder fine-tuning for coherent multi-speaker dubbing.
- [Online Demo](#).

ϵ ar-DiT

2025.11 – 2026.02

- Explored modern DiT optimization and representation-learning methods for audio, including REPA, JiT, DiTAR, and Lightning DiT+VAVAE, focusing on conditioning efficiency and audio representation quality.
- Investigated GAN-assisted DiT training for high-fidelity music reconstruction, reducing high-frequency over-smoothing and reconstruction artifacts.

Reathon

2025.12 – Present

- Built a Python library bridging Reaper DAW and Python-based deep learning environments, enabling session parsing, parameter mapping, and programmatic control for audio workflows and future audio-visual agents.
- **Online Documentation.**

EAR Audio Preview

2026.05 – Present

- Developed a VS Code extension for time-frequency audio analysis and monitoring, including spectrogram, stereo/polar analysis, LUFS metering, monitoring correction, and accelerated rendering.
- **Repository.**

Harmonic Transfer Envelope Algorithm

2024.02 – 2024.04

- Developed a lightweight neural nonlinear-processing module for compressor/plugin prototyping, connecting learned harmonic processing with practical music-production objectives.

EXPERIENCE

Qwen-omni Base Team | **Tech Report**, Music Gen Algorithm / Agentic Post-Training Engineer

2026.03 – Present

- **Core contributor / second author** to Qwen-Music, focusing on decoder-side audio quality, interactive mixing/editing capabilities, music evaluation, and data pipelines.
- Developed music-specific quality criteria and multimodal workflows connecting music understanding, reasoning, coding/function-calling, and data collection.

Initi AI, Ltd. | **yinchaoyongxian.com**, Music Generation Algorithm Engineer

2025.05 – 2026.03

- Contributed to the “Yinchao” large-scale music generation model, working on training loops and data-pipeline components with emphasis on mixing quality and overall audio fidelity.
- Conducted systematic listening and mixing evaluations on commercial tracks and generated music, feeding production-oriented observations back into model development.

CereProc Ltd | **cereproc.com**, TTS Developer

2024.11 – 2025.05

- Cleaned Chinese speech datasets and improved text-normalization pipelines; contributed to an LLM-based TTS system, including vocoder-side fine-tuning, SDK debugging, and feature development.

Xiamen Kazami Technology Co., Ltd. | **kazami.tech**, Mixing Engineer & Audio Developer

2023.09 – Present

- **3+ years of hands-on stem/live mixing and audio engineering**, developing creator-facing audio workflows and real-time production systems.
- Developed custom audio-control and routing software around Reaper using Lua and Python, and optimized real-time effects and production workflows for streamed events and content creators.
- Worked with production partners and platforms including Bilibili, Elephant Goose Media, Yuehua Entertainment, VirtuaReal, and NeoNova.
- Received professional mixing training at **Modern Sky**, one of China’s leading independent music companies.

COMMUNITY & LEADERSHIP

- Co-founded the academic collective **Ear-lab / Eps-Acoustic-Revolution-Lab**, serving as a key researcher and operator.
- Active open-source contributor; independently engineered **ReaperDoc** and related audio-production tooling.
- **Led the production and co-developed** “Elf of Era Idols Project”, an interactive fiction game achieving a **98% Positive Rating on Steam**.